Data Analysis Report: The Impact of Mental Health on Academic Productivity

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1. Executive Summary

This report aims to analyze the relationship between students' mental health and their academic productivity using Python data analysis techniques. The analysis relied on a dataset of 5,000 students, focusing on two main variables: mental health score (mental_health_score) and productivity score (productivity_score).

The results showed a significant positive correlation between the two variables, with a correlation coefficient of 0.63. The linear regression model revealed that each one-point increase in mental health leads to an average increase of 3.7 points in productivity. The model explains approximately 40% of the variance in productivity ($R^2 = 0.398$), indicating that mental health is a substantial influencing factor, while other factors contribute to the remaining variance.

The study recommends that educational institutions invest in student mental health support programs, given their positive impact on academic performance and productivity.

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2. Introduction

2.1 Problem Background

Recent studies indicate growing interest in students' mental health and its impact on their academic performance. However, the quantitative relationship between these two variables still requires further analysis using objective statistical tools.

2.2 Analysis Objectives

· Measure the strength of the relationship between students' mental health and productivity.

· Build a predictive model to estimate productivity based on mental health scores.

· Provide actionable recommendations for educational institutions.

2.3 Scope of Work

The analysis covers data from 5,000 students, focusing on mental health and productivity variables, in addition to analyzing the relationship across different academic levels.

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3. Methodology

3.1 Data Source

A CSV file containing 5,000 records and 21 columns was used. The main variables included in the analysis:

· mental_health_score: Mental health measure (value range: 1-10)

· productivity_score: Academic productivity measure (value range: 1-98)

· academic_level: Educational level (High School, Undergraduate, Postgraduate)

3.2 Analysis Tools

The following Python libraries were utilized:

· Pandas: Data processing and analysis

· Matplotlib & Seaborn: Data visualization

· Statsmodels: Regression modeling and result analysis

3.3 Analysis Steps

1. Data exploration and descriptive statistics

2. Correlation coefficient calculation

3. Visual exploration of relationships

4. Building a simple linear regression model

5. Interpreting results and drawing conclusions

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4. Data Exploration

4.1 Descriptive Statistics

Using the df.describe() function, the following statistics were obtained for the key variables:

Table 1: Descriptive Statistics for Mental Health and Productivity

Variable	Count	Mean	Std Dev	Min	Max
Mental Health	5000	5.51	2.87	1.00	10.00
Productivity	5000	37.27	16.85	1.00	98.02

4.2 Correlation Coefficient

Pearson correlation coefficient was calculated between the two variables:

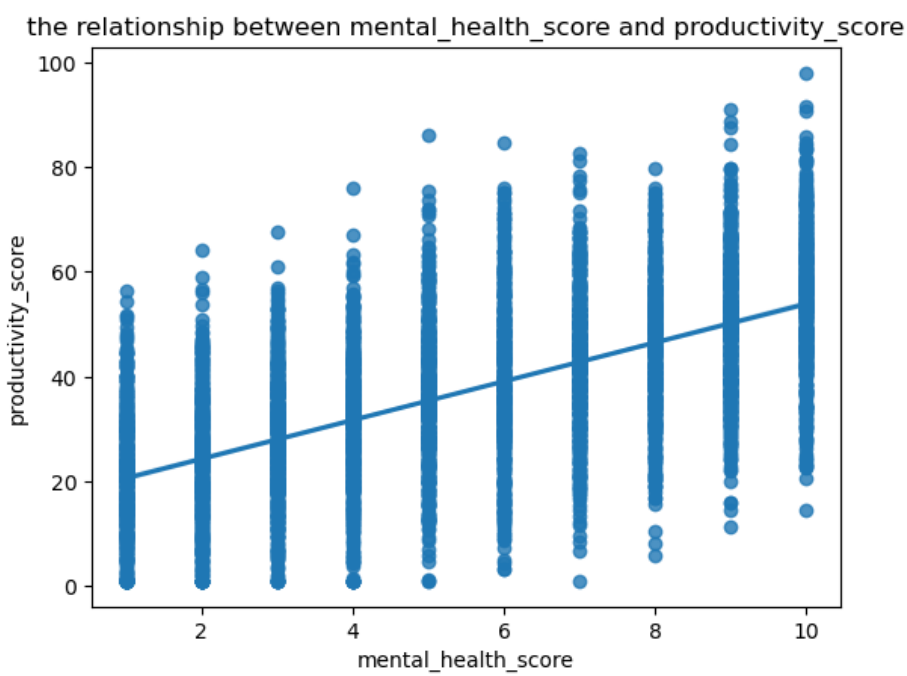
```
df[['mental_health_score', 'productivity_score']].corr()
```

Result: Correlation coefficient = 0.63

Statistical Interpretation: The value 0.63 indicates a moderate to strong positive correlation between mental health and productivity, meaning that an increase in mental health is associated with a noticeable increase in productivity.

4.3 Visualization

Figure 1: General Relationship Between Mental Health and Productivity



regplot was used to visualize the relationship with a regression line:

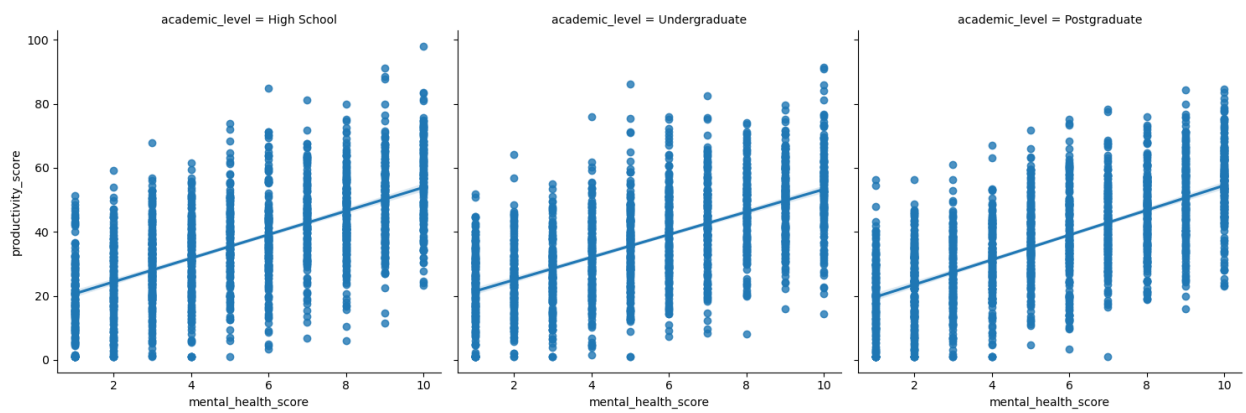
```
sns.regplot(x='mental_health_score', y='productivity_score',
data=df)

plt.title('the relationship between mental_health_score and
productivity_score')

plt.show()
```

The plot shows data points clustered around the regression line, confirming the numerical correlation result.

Figure 2: Relationship by Academic Level



lmplot was used to analyze the relationship across different academic levels:

```
sns.lmplot(x='mental_health_score', y='productivity_score',

           data=df, col='academic_level',

           col_order=['High School', 'Undergraduate',
'Postgraduate'])

plt.show()
```

Results showed that the positive relationship between mental health and productivity is consistent across all academic levels, enhancing the generalizability of the findings.

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5. Linear Regression Model

5.1 Model Building

A simple linear regression model was built using the Statsmodels library, where:

· Dependent variable (Y): Productivity (productivity_score)

· Independent variable (X): Mental health (mental_health_score)

```
import statsmodels.api as sm
X = df['mental_health_score']
y = df['productivity_score']
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
print(model.summary())
```

5.2 Model Results

Table 2: Regression Model Outputs

Variable	Coefficient	Std Error	P-value
Intercept	16.87	0.40	0.000
Mental Health	3.70	0.064	0.000

Model Quality Indicators:

· R-squared: 0.398

· F-statistic: 3300 (Prob = 0.00)

5.3 Interpretation of Results

Regression Equation

The relationship can be expressed as:

Predicted Productivity = 16.87 + (3.70 × Mental Health Score)

Quantitative Impact

· Each one-point increase in mental health score leads to an average increase of 3.7 points in productivity, holding other factors constant.

Statistical Significance

· P-value < 0.05, confirming that the relationship is statistically significant and not due to chance.

· The large F-statistic (3300) with P-value = 0.00 confirms that the model as a whole is statistically significant.

Explanatory Power

· R-squared = 0.398 means that approximately 40% of the variance in productivity can be explained by mental health alone.

· The remaining 60% is attributed to other factors not included in the model (e.g., study hours, sleep, stress, social support), opening the door for more comprehensive future studies.

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6. Conclusions

1. Positive and Significant Relationship: There is a statistically significant positive relationship between mental health and academic productivity. Students with better mental health achieve higher productivity.

2. Predictability: The estimated regression model can be used to predict productivity based on mental health scores. For example, a student with a mental health score of 7 is expected to have a productivity score of 42.8.

3. Consistency Across Academic Levels: The relationship between the variables exists with similar strength across all educational stages (High School, Undergraduate, Postgraduate).

4. Limitations of Interpretation: The mental health model alone explains 40% of the variance in productivity, indicating the importance of studying other variables for a better understanding of influencing factors.

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7. Recommendations

For Educational Institutions:

1. Investment in Mental Health Support Programs: Given the clear positive impact of mental health on productivity, it is recommended to allocate resources to psychological support programs and student counseling.

2. Early Intervention: The model can be used as an initial screening tool to identify students most at risk of low productivity based on their mental health assessment.

3. Designing Supportive Environments: Work towards creating an academic environment that promotes mental health by reducing stress and providing social support.

For Future Studies:

1. Model Expansion: Adding other variables such as study_hours, sleep_hours, stress_level to improve the model's explanatory power.

2. Causal Analysis: Using advanced statistical techniques to examine causal relationships rather than correlational ones.

3. Application in Diverse Contexts: Replicating the study on samples from different countries and disciplines to test the generalizability of results.

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8. References

· McKinney, W. (2017). Python for Data Analysis (2nd ed.). O'Reilly Media.

· Pandas Development Team. (2025). pandas documentation. Retrieved from https://pandas.pydata.org/docs/

· Seaborn Development Team. (2025). seaborn: statistical data visualization. Retrieved from https://seaborn.pydata.org/

· Statsmodels Development Team. (2025). statsmodels: statistics in Python. Retrieved from https://www.statsmodels.org/

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9. Appendices

Appendix A: Complete Code

```
# Import libraries

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

import statsmodels.api as sm

# Read data

df =
pd.read_csv('ultimate_student_productivity_dataset_5000.csv')

# Data exploration

print(df.info())

print(df.describe())

# Correlation coefficient

print(df[['mental_health_score', 'productivity_score']].corr())

# Relationship plot

sns.regplot(x='mental_health_score', y='productivity_score',
data=df)

plt.title('The Relationship Between Mental Health and
Productivity')

plt.show()

# Plot by academic level

sns.lmplot(x='mental_health_score', y='productivity_score',

           data=df, col='academic_level',

           col_order=['High School', 'Undergraduate',
'Postgraduate'])

plt.show()

# Regression model

X = df['mental_health_score']

y = df['productivity_score']

X = sm.add_constant(X)
```

```
model = sm.OLS(y, X).fit()
print(model.summary())
```

Appendix B: Definition of Key Variables

Variable	Type	Description
mental_health_score	float (10-1)	Mental health measure (1-10)
productivity_score	float (98-1)	Academic productivity measure (1-98)
academic_level	categorical	Educational level